Cryptography

* cryptography is the art and science of securing digital information through mathematical techniques.

Security layers

* computer security
* Network security
* internet security
* cyber security

→ plain text
The original intelligible message
→ cipher text
The transformed message
→ key
information used by the cipher, known only to the sender & receiver.
→ cipher
An algorithm for transforming, by transposition or
→ Encipher
converting plaintext to cipher text using a cipher and a key.
→ Decipher
Converting cipher text to plaintext using a cipher and a key

Cryptography

Classified into three independent dimentions

i) Type of operations for tran

- Substitution
  - element in plain text is mapped into another element
- transformation.
  - element in the plain text are rearranged.

ii) The number of keys used

- the sender and receiver uses same key
- symmetric key or single key encryption.
- the sender and receiver uses different key

public key encryption

iii) The way in which the plain text is processed

A block cipher.

- process the input and block of elements at

a time.

- producing output block for each input block

A stream cipher

- process the input element continuosly
- producing output element one at a time

Aspects of security

There are four general categories of attack:

1. interruption - An asset of the system is destroyed or unavai
2. interception - An unauthorized party gains
3. Modification - when unauthorized party gains acess & tampers
4. fabrication -

14/01/26

Caesar Cipher

Encrypt the following plain text using the Caesar cipher with a shift key of 3.

plain text

CYBER SECURITY

cipher text

FBEHUV HF XULWB

→ Recovery

Encrypt the following cipher with a shift key of 3

Plain text

CYBER SECURITY

• shift each letter

*

If two letters are in the same row, then replace with immediate right letters.

Plain text - cd

p	l	a	y	f
i/j	r	e	x	m
b	c	d	g	h
k	n	o	q	s
t	u	v	w	z

Cipher text - dg

p	l	a	y	f
i/j	r	e	x	m
b	c	d	g	h
k	n	o	q	s
t	u	v	w	z

* plain text : computer
key : security

⇒ co / mp / ut / er

S	E	C	U	R
I/J	T	Y	A	B
D	F	G	H	K
L	M	N	O	P
Q	V	W	X	Z

CO - UN
mp - NL
UT - AE
ER - CS

Hill cipher

- Hill cipher is a classical polygraphic substitution cipher.
- It uses matrix m

HELLO :

H - 7
E - 4
L - 11
L - 11
O - 14
X - 23

key = 2x2

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

HELLO → HE | LL | OX

Encrypt Each block

i) HE $= (7, 4)$

$$= \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 7 \\ 4 \end{pmatrix} = \begin{bmatrix} 33 \\ 34 \end{bmatrix}$$

Mod 26

33 mod 26 = 7 → H
34 mod 26 = 8 → i

ii) L L $= (11, 11)$

$$= \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 11 \\ 11 \end{pmatrix} = \begin{bmatrix} 66 \\ 77 \end{bmatrix}$$

66 mod 26 = 14 ⇒ O
77 mod 26 = 25 → Z

iii) O X $= [14, 23]$

$$= \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 14 \\ 23 \end{pmatrix} = \begin{bmatrix} 111 \\ 143 \end{bmatrix}$$

111 mod 26 = 7 → H
143 mod 26 = 13 → N

Cipher Text = HELLO → HIOZHN

Decryption:

$P = k^{-1} \times C \pmod{26}$

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \Rightarrow 3 \times 5 - 3 \times 2 = 9$$
$$\Rightarrow 9^{-1} = 3$$

$$k^{-1} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

i) HI → (7, 8)

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 241 \\ 212 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix} \text{mod } 26$$

HI = HE

ii) OZ

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 14 \\ 25 \end{bmatrix} = \begin{bmatrix} 635 \\ 505 \end{bmatrix} = \begin{bmatrix} 11 \\ 11 \end{bmatrix} \text{mod } 26$$

OZ = LL

iii) HN

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 13 \end{bmatrix} = \begin{bmatrix} 326 \\ 257 \end{bmatrix} = \begin{bmatrix} 14 \\ 23 \end{bmatrix} \text{mod } 26$$

HN = OX

plain text = HIOZHN → HELLO